GNANESHKANTHAN S

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Embedded Systems - IoT - Electronics Engineer

Career Objective

Aspiring Embedded Systems Designer with a strong interest in microcontrollers, Embedded C, and IoT applications. Eager to apply technical knowledge and gain practical experience through real-world engineering projects while continuously developing professional and technical skills.

Education

B.E - ECE 2023 – 2027 GCE, Erode CGPA: 7.89

Higher Secondary Education Vivekananda Vidhyalaya Matriculation Higher Secondary School Percentage: 93%

Technical Skills

- **Programming Languages:** C, Python(Basics)
- **Embedded Systems:** Arduino, ESP32 ,Raspberry PI
- **Design & Simulation Tools:** MATLAB, Multisim, Proteus , Keil
- **Development Environment:** Arduino IDE
- **Virtual Simulation Platforms:** Wokwi, Tinkercad
- **Communication Protocols:** UART, SPI, I2C

Projects

Fire Detection System using Raspberry Pi

- Developed an intelligent fire detection system using Raspberry Pi and computer vision techniques.
- Trained and deployed a machine learning model to detect fire and smoke from real-time camera feeds.
- Implemented real-time monitoring and alert generation for early fire detection.
- Optimized image processing and inference for embedded deployment on Raspberry Pi.

Tools: Raspberry Pi, Python, OpenCV, TensorFlow Lite, Camera Module

Smart Irrigation System for Precision Farming

- Developed an IoT-based irrigation system using sensors and control units.
- Automated irrigation based on soil moisture, crop requirements, and environmental conditions.
- Implemented real-time monitoring to reduce water wastage and improve crop productivity.
- Enhanced agricultural efficiency through sensor-driven automation and smart decision-making.

Tools: Arduino, IoT, Sensors, Embedded Systems

Internship

Embedded Systems and C Programming Intern 30 Days
Pumo Technovation India Pvt. Ltd.

- Gained hands-on experience in microcontroller interfacing, circuit simulation, and embedded software development.
- Developed and simulated embedded system circuits using Proteus Design Suite.
- Worked with Keil uVision IDE for program development, compilation, and debugging.
- Strengthened problem-solving and debugging skills through practical embedded system exercises.

Certifications

- **Introduction to Embedded System Design** NPTEL
- **Embedded C and Microcontroller Programming** Naan Mudhalvan
- **Industrial IoT and Industry 4.0** Naan Mudhalvan
- **Digital Circuits** NPTEL
- **Embedded for Beginners** NIELIT Calicut
- **Introduction to IoT** Infosys Springboard

Workshops

- MATLAB Workshop conducted at Government College of Engineering, Erode.
- 5-Day PCB Design Workshop conducted at Government College of Engineering, Erode.
- Robotics Workshop conducted at Government College of Engineering, Erode.

Soft Skills

Teamwork and Collaboration, Time Management, Critical Thinking, Attention to Detail